Ch 9.3-4: Support Vector Machine

Lecture 27 - CMSE 381

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Announcements

Last time:

- 9.2 Support Vector Classifier

This lecture:

- 9.3 Support Vector Machine

Announcements:

20	F	Nov 4	Polynomial & Step Functions.	7.1,7.2	
21	M	Nov 7	Step Functions	7.2	
22	W	Nov 9	Basis functions, Regression Splines	7.3,7.4	
23	F	Nov 11	Decision Trees	8.1	HW #7 Due
24	M	Nov 14	Random Forests	8.2.1, 8.2.2	
25	W	Nov 16	Maximal Margin Classifier	9.1	
26	F	Nov 18	SVC	9.2	HW #8 Due
27	M	Nov 21	SVM	9.3, 9.4, ~~9.5~~	
28	W	Nov 23	Extended virtual office hours		
	F	Nov 25	No class - Thanksgiving		
29	M	Nov 28	Single layer NN	10.1	HW #9 Due
30	W	Nov 30	Multi Layer NN	10.2	
31	F	Dec 2	CNN	10.3	
32	M	Dec 5	Unsupervised Learning & Clustering	12.1, 12.4	HW #10 Due
	W	Dec 7	Review		
	F	Dec 9	Midterm #3	Bring your cheat sheet and a non-internet-connected calculator	

Section 1

Last Time

Classification Setup

Data matrix:

$$X = \begin{pmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{pmatrix}_{n \times p}$$

Observations in one of two classes,

$$y_i \in \{-1, 1\}$$

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$$x_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{1p} \end{pmatrix}, \cdots, x_n = \begin{pmatrix} x_{n1} \\ \vdots \\ x_{np} \end{pmatrix}$$

Separate out a test observation

$$x^* = (x_1^* \ \cdots \ x_p^*)^T$$

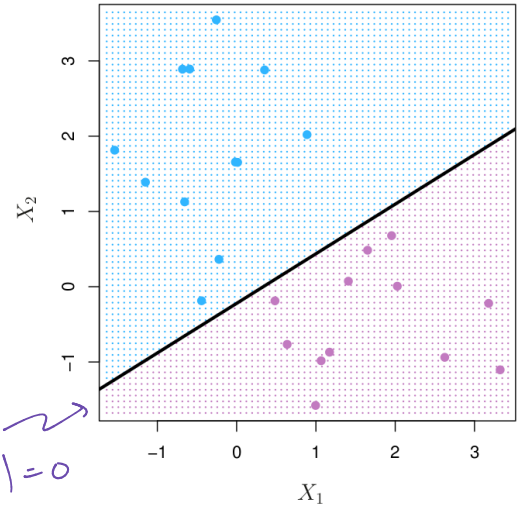
Hyperplane becomes a classifier

If you have a separating hyperplane:

- Check

$$f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$$

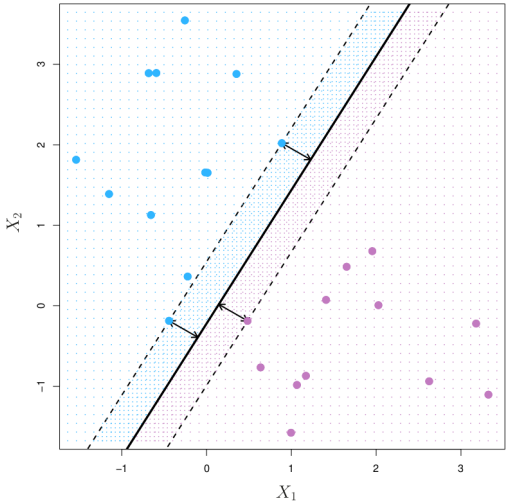
- If positive, assign $\hat{y} = 1$
- If negative, assign $\hat{y} = -1$



$f(x) = 0$

How do we pick? Old version

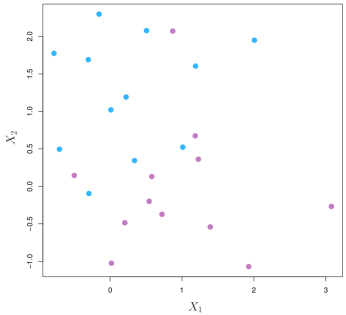
Maximal margin classifier



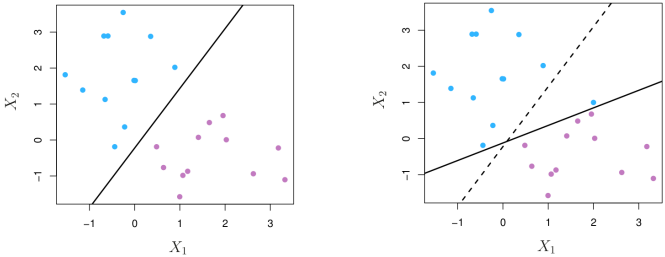
Required Separating hyperplanes

- For a hyperplane, the *margin* is the smallest distance from any data point to the hyperplane.
- Observations that are closest are called *support vectors*.
- The *maximal margin hyperplane* is the hyperplane with the largest margin
- The classifier built from this hyperplane is the *maximal margin classifier*.

Issues

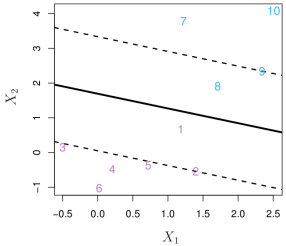


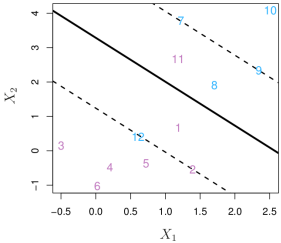
No separating hyperplane exists



Choice of hyperplane is sensitive to new points

Support Vector Classifier → Soft margin





$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq \underline{\underline{C}},$$

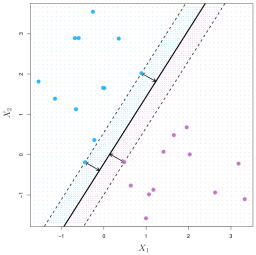
Two formulations side by side

Maximal Margin Classifier

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \; M$$

$$\text{subject to} \; \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \;\; \forall\, i = 1, \ldots, n$$



Support Vector Classifier

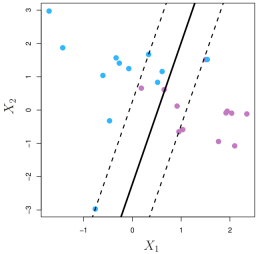
$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\, M}{\text{maximize}} \; M$$

$$\text{subject to} \; \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

$$\epsilon_i \geq 0, \;\; \sum_{i=1}^{n} \epsilon_i \leq C,$$

Control measure of violations



So many variables

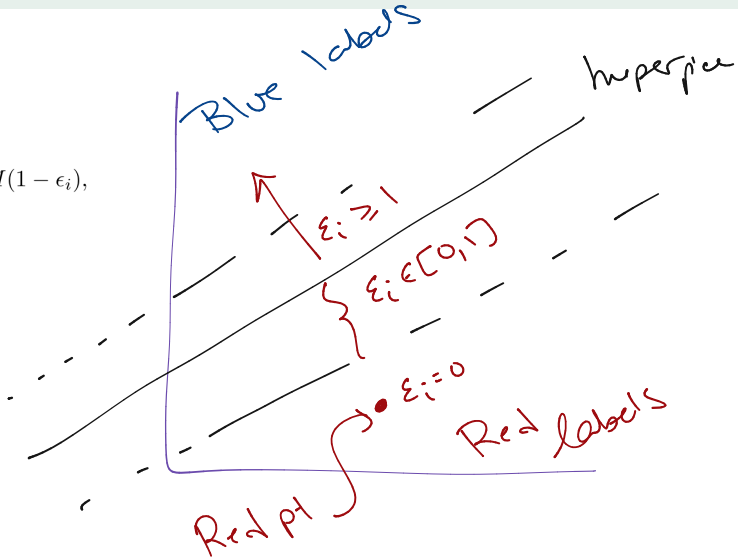
$$\underset{\beta_0,\beta_1,\ldots,\beta_p,\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

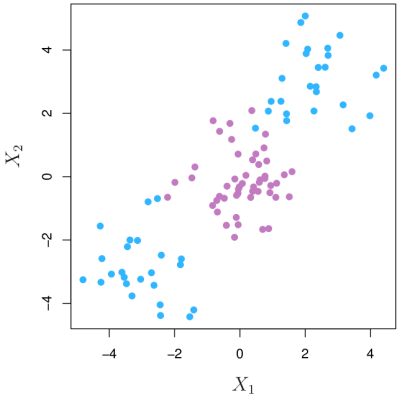
$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M(1 - \epsilon_i),$$

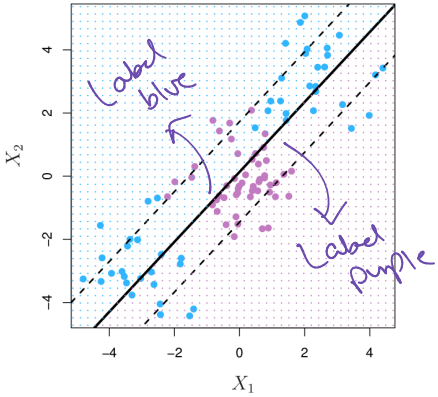
$$\epsilon_i \geq 0, \quad \sum_{i=1}^{n} \epsilon_i \leq C,$$

Not the same C as in the code, but controls the same thing



Limiting factor of SVC





Section 2

Support Vector Machine

Example of using more features

Pro: Flexibility

Want $2p$ features:

$$X_1, X_1^2, X_2, X_2^2, \cdots, X_p, X_p^2$$

Problem:
lots of features
$X_i X_j$

Optimization becomes:

$$\underset{\beta_0,\beta_{11},\beta_{12},\ldots,\beta_{p1},\beta_{p2},\epsilon_1,\ldots,\epsilon_n,\,M}{\text{maximize}} \quad M$$

$$\text{subject to } y_i\left(\beta_0 + \sum_{j=1}^{p}\beta_{j1}x_{ij} + \sum_{j=1}^{p}\beta_{j2}x_{ij}^2\right) \geq M(1-\epsilon_i),$$

$$\sum_{i=1}^{n}\epsilon_i \leq C, \quad \epsilon_i \geq 0, \quad \sum_{j=1}^{p}\sum_{k=1}^{2}\beta_{jk}^2 = 1.$$

Kernels

Idea: Enlarge the feature space

But

Want a computationally efficient manner w/o just duplicating lots of variables

Inner products

$$\langle a, b \rangle = \sum_{i=1}^{r} a_i b_i$$

$a = (a_1, a_2, \ldots, a_r)$ $\quad$ $b = (b_1, b_2, \ldots, b_r)$

Quick computations

What are the following?

- $\langle (1,1),(0,3) \rangle = 1 \cdot 0 + 1 \cdot 3 = 3$
- $\langle (1,1),(3,2) \rangle = 1 \cdot 3 + 1 \cdot 2 = 5$
- $\langle (2,3),(0,3) \rangle = 2 \cdot 0 + 3 \cdot 3 = 9$
- $\langle (2,3),(3,2) \rangle = 2 \cdot 3 + 3 \cdot 2 = 12$

SVC via inner products

$\beta = (\beta_1, \beta_2, \ldots, \beta_p)$

$$f(x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p = \beta_0 + \langle \beta, x \rangle$$

$x = (x_1, x_2, \ldots, x_p)$

Magic:

There are always coefficients $\alpha_1, \ldots, \alpha_n$

for which

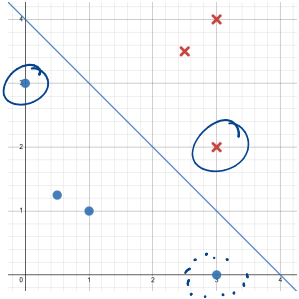
$$f(x) = \beta_0 + \sum_{i=1}^{n} \alpha_i \langle x, x_i \rangle$$

data points

Example

$$-2\sqrt{2} + \frac{\sqrt{2}}{2}X_1 + \frac{\sqrt{2}}{2}X_2 = 0$$

$$-f(x_1, x_2) = -2\sqrt{2} + \frac{\sqrt{2}}{18}\langle (X_1, X_2), (0,3)\rangle + \frac{\sqrt{2}}{6}\langle (X_1, X_2), (3,2)\rangle = 0$$



$$f(1,1) = -2\sqrt{2} + \frac{\sqrt{2}}{18}(3) + \frac{\sqrt{2}}{6}(5)$$

$$= -2\sqrt{2} + \frac{1}{6}\sqrt{2} + \frac{5}{6}\sqrt{2}$$

$$= -\sqrt{2}$$

$\Rightarrow$ predict $y = -1$

What are the α_is?

Data point	α_i
$(3,4)$	0
$(2.5,3.5)$	0
$\rightarrow (3,2)$	$\frac{\sqrt{2}}{6}$
$(3,0)$	0
$\rightarrow (0,3)$	$\frac{\sqrt{2}}{18}$
$(1,1)$	0
$(0.5,1.25)$	0

What α_i's are needed to write the hyperplane

$$\underbrace{-2\sqrt{2}}_{\beta_0} + \frac{\sqrt{2}}{18}\langle \underbrace{(X_1,X_2)}_{X}, \underbrace{(0,3)}\rangle + \frac{\sqrt{2}}{6}\langle (X_1,X_2), \underbrace{(3,2)}\rangle$$

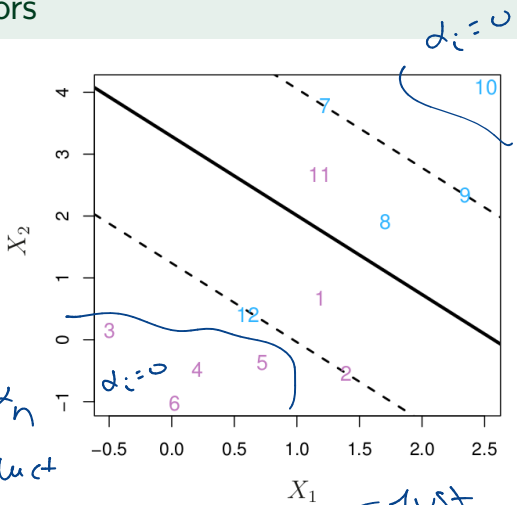
of the previous page in the form

$$f(x) = \beta_0 + \sum_{i\in\mathcal{S}} \alpha_i \langle x, x_i\rangle ?$$

$\alpha_i \langle x, (0,3)\rangle$

SVC via inner products of support vectors

α_i ends up non zero for only the support vectors



$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i \langle x, x_i \rangle$$

To estimate the parameters $\alpha_1, \ldots, \alpha_n$ need $\frac{n(n-1)}{2} = \binom{n}{2}$ inner product values

The point: Representing a linear classifier just needs inner products

The kernel

$$K(\underbrace{x_i, x_i'}_{\text{two input vectors / data pts}})$$

Could have a linear kernel

$$K(x_i, x_i') = \langle x_i, x_i' \rangle$$

Could have something else

Similarity measure

returns a number

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i)$$

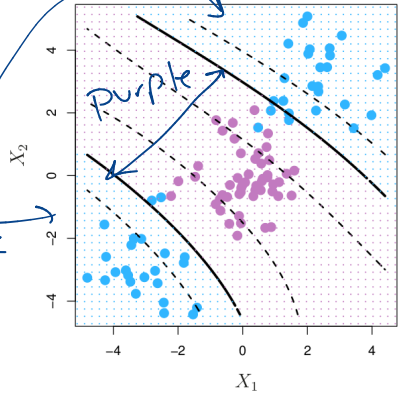
A support vector machine (SVM) is SVC from before but where K might not be linear kernel.

A polynomial kernel

$f(x) = \beta_0 + \sum \alpha_i K(x_i, x_i')$

$$K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij} x_{i'j}\right)^d$$

user decides d = degree



This example $d = 3$

A radial kernel

user chooses γ

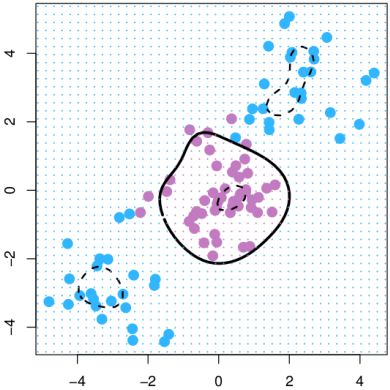
$$K(x_i, x_i') = \exp\left(-\gamma \underbrace{\sum_{j=1}^{p}(x_{ij} - x_{i'j})^2}\right)$$

Pts far away from each other

Distance Squared $\sum_{j=1}^{p}(x_{ij} - x_{i'j})^2$

Then $e^{-\gamma \cdot \text{big}} \Rightarrow$ small

$f(x)$ is rigged so that x_i plays little role in determining x if far away



Support Vector Machine

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i K(x, x_i)$$

Learn

- $\alpha_i > 0 \rightsquigarrow x_i$ is support vector
- Pick some kernel K (plus its inputs) in advance

Coding

Section 3

SVM with more than two classes

One-Vs-One Classification

Also called all-pairs

qualitative

Predict y_i with values $\{1, \ldots, K\}$

Construct $\binom{K}{2}$ SVMs each of which compares a pair of classes

Too many models!

predict label most frequent for a given data point.

One-Vs-All Classification

$y_i \in \{1, \ldots, K\}$

Fit K diff models

each one predicts label i -or- not label i

Model $_1$ $\rightsquigarrow$ predict if data point is apple or not an apple

Model $_2$ $\rightsquigarrow$ predict if — " — orange or not orange

Output for each model

$$\left.\begin{array}{l} f_1(x_*) \\ \vdots \\ f_K(x_*) \end{array}\right\} \text{pick the label w/ model having largest function value}$$

farthest from the boundary

TL;DR

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \alpha_i \langle x, x_i \rangle$$

Kernels

- Linear

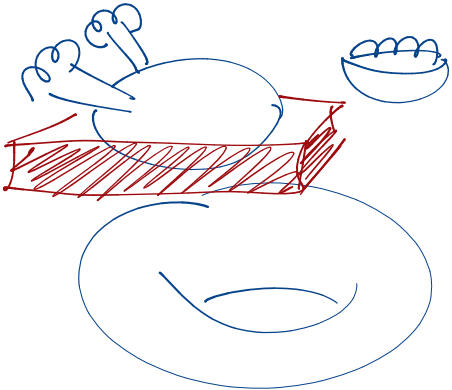
$$K(x_i, x_{i'}) = \sum_{j=1}^{p} x_{ij} x_{i'j}$$

- Polynomial

$$K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij} x_{i'j}\right)^d$$

- Radial

$$K(x_i, x_i') = \exp\left(-\gamma \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2\right)$$

Next time

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